

SHANTOU, China

WMO: 593160

Lat: 23.3853N

Lon: 116.6792E

Elev: 10

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	45.1	47.9	29.9	23.9	54.7	35.1	30.0	53.9	19.9	58.2	17.9	58.1	4.8	20	0.341

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	94.8	80.6	93.0	80.6	91.2	80.4	83.6	89.2	82.7	88.0	81.9	87.1	8.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.4	169.0	86.7	81.0	161.1	85.3	80.7	159.4	85.0	47.8	89.4	46.8	88.6	46.0	87.9	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.0	15.6	13.5	DB	40.3	98.3	3.0	1.7	38.2	99.6	36.4	100.6	34.7	101.5	32.5	102.8
			WB	36.0	85.3	4.2	1.3	33.0	86.3	30.5	87.1	28.2	87.8	25.2	88.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.2	59.2	60.3	64.3	71.6	77.9	82.5	84.7	84.4	82.4	77.2	70.2	62.3
	DBStd	10.37	5.26	6.09	5.82	5.64	4.53	3.40	2.75	2.77	3.04	3.76	4.87	5.28
	HDD50	7	3	3	0	0	0	0	0	0	0	0	0	1
	HDD65	566	191	153	84	12	0	0	0	0	0	0	12	114
	CDD50	8458	287	291	444	649	864	976	1075	1068	973	842	607	382
	CDD65	3541	10	21	63	210	399	526	610	603	523	377	169	30
	CDH74	34244	22	63	218	1057	3127	5745	7520	7236	5723	2849	633	51
	CDH80	12382	1	5	30	236	869	2068	3284	3043	2089	675	80	2
(14) Wind	WSAvg	5.9	5.8	6.1	6.1	5.8	5.6	5.9	5.9	5.7	5.8	6.3	5.9	5.7
(15) Precipitation	PrecAvg	61.3	1.3	2.0	3.3	5.9	7.8	11.8	8.4	9.7	5.8	1.9	1.7	1.3
	PrecMax	98.8	9.8	12.6	16.3	23.7	23.3	28.4	19.1	25.6	21.7	9.9	9.2	5.2
	PrecMin	36.5	0.0	0.0	0.2	0.1	0.3	1.4	1.5	1.1	0.6	0.0	0.0	0.0
	PrecStd	14.5	1.5	2.6	3.3	4.6	4.8	6.1	4.8	6.1	4.1	2.6	2.0	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.0	80.4	83.9	88.2	93.4	95.1	96.9	96.7	94.9	91.3	85.4	78.9
		MCWB	66.5	69.9	73.1	76.7	80.5	80.4	81.1	81.0	79.4	76.7	73.5	68.5
	2%	DB	71.9	75.4	79.2	85.6	89.5	92.2	94.7	94.2	91.8	87.8	82.3	75.0
		MCWB	63.3	67.6	70.8	75.9	79.8	81.0	81.1	80.7	79.4	76.0	72.2	66.3
	5%	DB	68.7	71.6	76.0	82.7	87.3	89.9	92.4	91.9	89.9	85.8	80.2	72.0
		MCWB	61.0	64.9	68.7	74.7	78.7	80.6	80.9	80.7	78.6	74.6	70.7	64.0
	10%	DB	66.4	68.2	73.0	80.2	84.5	88.1	90.5	90.0	88.2	83.9	77.3	70.0
		MCWB	60.0	62.6	66.5	73.6	77.4	80.5	80.6	80.3	78.1	73.5	69.3	62.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.3	72.9	75.0	79.3	82.8	84.1	84.7	84.6	83.4	79.8	76.5	71.4
		MCDB	74.5	75.7	80.7	84.3	87.5	88.8	91.2	90.7	89.0	85.6	81.3	75.8
	2%	WB	65.4	69.1	72.7	77.6	81.2	82.8	83.5	83.4	81.9	78.6	74.6	68.7
		MCDB	70.0	74.0	77.6	82.6	86.3	87.4	89.5	89.2	87.4	84.0	79.2	72.8
	5%	WB	63.2	66.2	70.4	76.1	79.9	82.0	82.5	82.4	80.8	77.5	73.1	66.7
		MCDB	67.3	70.1	74.2	81.0	84.9	86.6	88.1	87.9	86.2	82.7	77.6	70.7
	10%	WB	61.2	63.7	68.0	74.6	78.8	81.2	81.7	81.5	80.0	76.2	71.4	64.6
		MCDB	65.2	67.7	71.7	79.0	83.2	85.9	87.2	87.0	85.1	81.3	76.1	68.8
(35) Mean Daily Temperature Range	5% DB	MDBR	12.6	11.9	12.1	11.5	10.7	10.1	11.3	11.5	12.0	12.9	13.4	13.5
		MCDBR	15.8	17.1	17.2	15.1	14.0	12.8	14.2	14.3	14.0	14.4	14.8	15.1
		MCWBR	8.3	9.6	9.6	7.5	6.5	5.1	5.3	5.6	5.7	5.9	6.9	7.9
	5% WB	MCDBR	13.5	15.4	15.3	13.8	12.5	11.0	12.0	12.3	12.2	11.8	12.6	12.5
		MCWBR	8.1	10.1	10.4	7.8	6.8	5.6	5.9	6.0	5.7	5.8	6.8	7.5
(40) Clear-Sky Solar Irradiance	taub		0.568	0.639	0.755	0.743	0.599	0.557	0.536	0.597	0.638	0.635	0.560	0.553
	taud		1.807	1.700	1.520	1.585	1.918	2.038	2.094	1.909	1.792	1.742	1.895	1.863
	Ebn at Noon		217	211	195	202	233	240	246	231	218	209	218	214
	Edh at Noon		62	73	92	87	62	55	52	62	69	69	56	57
(44) All-Sky Solar Radiation	RadAvg		979	1023	1120	1208	1381	1451	1748	1614	1522	1373	1094	993
	RadStd		140	173	154	194	196	193	141	151	108	155	122	132

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.61	N/A	N/A	+1.62	N/A	N/A	N/A	N/A	+248	+176
(47) Regional (4 neighbors)		+0.49	N/A	N/A	+0.68	N/A	N/A	N/A	N/A	+153	+142

Nomenclature: See separate page